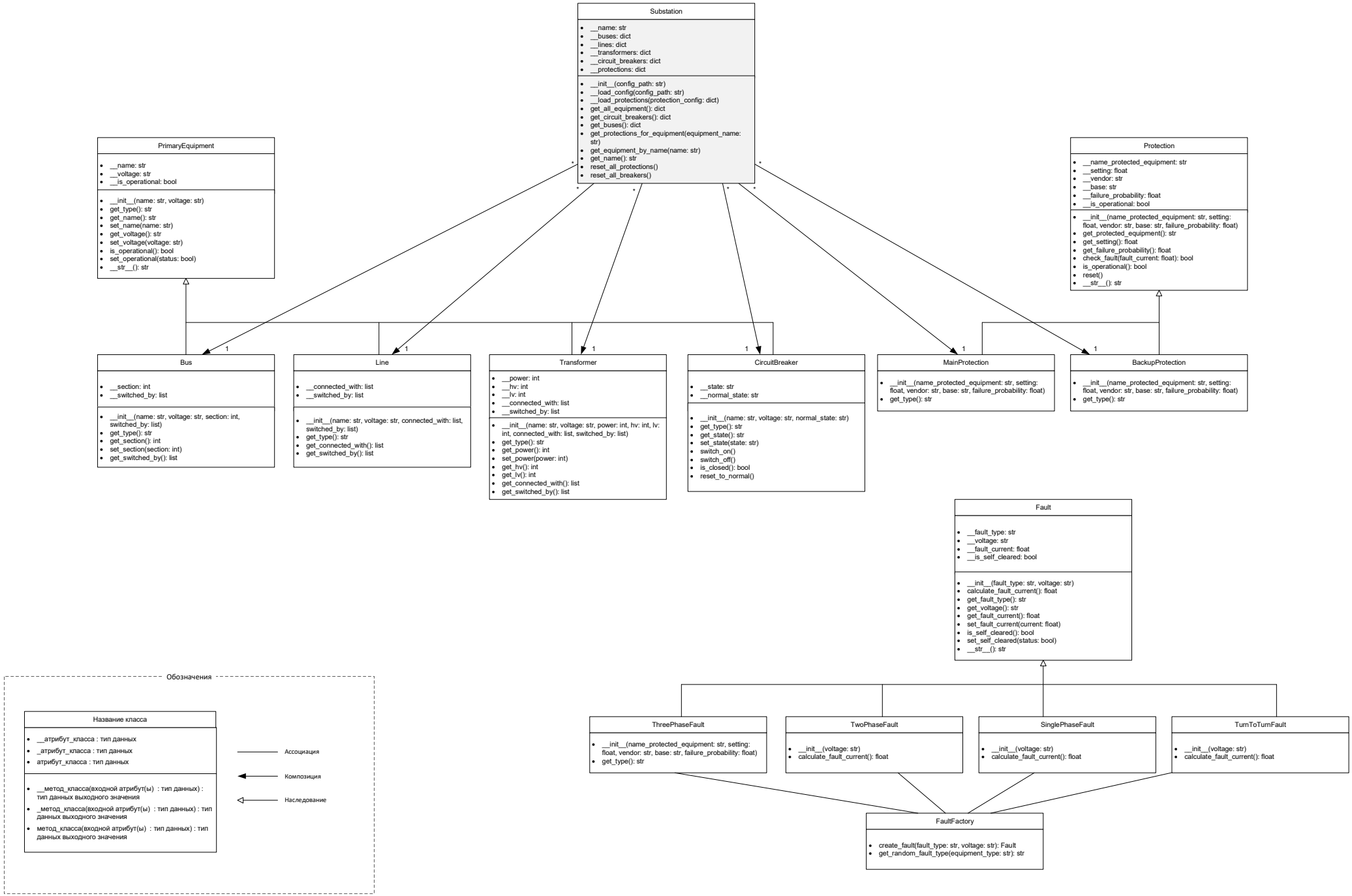
Substation
__name: str
__buses: dict
__lines: dict
__transformers: dict
__circuit_breakers: dict
__protections: dict
__init__(config_path: str)
__load_config(config_path: str)
__load_protections(protection_config: dict)
get_all_equipment(): dict
get_circuit_breakers(): dict
get_buses(): dict
get_protections_for_equipment(equipment_name: str)
get_equipment_by_name(name: str)
get_name(): str
reset_all_protections()
reset_all_breakers()
PrimaryEquipment
__name: str
__voltage: str
__is_operational: bool
__init__(name: str, voltage: str)
get_type(): str
get_name(): str
set_name(name: str)
get_voltage(): str
set_voltage(voltage: str)
is_operational(): bool
set_operational(status: bool)
__str__(): str
Protection
__name_protected_equipment: str
__setting: float
__vendor: str
__base: str
__failure_probability: float
__is_operational: bool
__init__(name_protected_equipment: str, setting: float, vendor: str, base: str, failure_probability: float)
get_protected_equipment(): str
get_setting(): float
get_failure_probability(): float
check_fault(fault_current: float): bool
is_operational(): bool
reset()
__str__(): str
Bus
__section: int
__switched_by: list
__init__(name: str, voltage: str, section: int, switched_by: list)
get_type(): str
get_section(): int
set_section(section: int)
get_switched_by(): list
Line
__connected_with: list
__switched_by: list
__init__(name: str, voltage: str, connected_with: list, switched_by: list)
get_type(): str
get_connected_with(): list
get_switched_by(): list
Transformer
__power: int
__hv: int
__lv: int
__connected_with: list
__switched_by: list
__init__(name: str, voltage: str, power: int, hv: int, lv: int, connected_with: list, switched_by: list)
get_type(): str
get_power(): int
set_power(power: int)
get_hv(): int
get_lv(): int
get_connected_with(): list
get_switched_by(): list
CircuitBreaker
__state: str
__normal_state: str
__init__(name: str, voltage: str, normal_state: str)
get_type(): str
get_state(): str
set_state(state: str)
switch_on()
switch_off()
is_closed(): bool
reset_to_normal()
MainProtection
__init__(name_protected_equipment: str, setting: float, vendor: str, base: str, failure_probability: float)
get_type(): str
BackupProtection
__init__(name_protected_equipment: str, setting: float, vendor: str, base: str, failure_probability: float)
get_type(): str
Fault
__fault_type: str
__voltage: str
__fault_current: float
__is_self_cleared: bool
__init__(fault_type: str, voltage: str)
calculate_fault_current(): float
get_fault_type(): str
get_voltage(): str
get_fault_current(): float
set_fault_current(current: float)
is_self_cleared(): bool
set_self_cleared(status: bool)
__str__(): str
ThreePhaseFault
__init__(name_protected_equipment: str, setting: float, vendor: str, base: str, failure_probability: float)
get_type(): str
TwoPhaseFault
__init__(voltage: str)
calculate_fault_current(): float
SinglePhaseFault
__init__(voltage: str)
calculate_fault_current(): float
TurnToTurnFault
__init__(voltage: str)
calculate_fault_current(): float
FaultFactory
create_fault(fault_type: str, voltage: str): Fault
get_random_fault_type(equipment_type: str): str
Обозначения
Название класса
__атрибут_класса : тип данных
_атрибут_класса : тип данных
атрибут_класса : тип данных
__метод_класса(входной атрибут(ы) : тип данных) : тип данных выходного значения
_метод_класса(входной атрибут(ы) : тип данных) : тип данных выходного значения
метод_класса(входной атрибут(ы) : тип данных) : тип данных выходного значения
Ассоциация
Композиция
Наследование